Lab 5: Ethernet Frames and ARP Analysis using Wireshark

Assignment Overview: In this lab, you will use Wireshark to capture and analyze Ethernet frames and ARP packets, gaining a deeper understanding of their structure and role in network communication. You'll also explore protocol encapsulation, MAC address manufacturer information, and ARP response timing.

Assignment Instructions:

1- **Explanation Requirement:** Briefly justify your approach and answers in a concise manner.

2- **Screenshots:** Include relevant screenshots of your Wireshark results as evidence for each answer.

3- **Detail Orientation:** Pay attention to packet details, header fields, and overall protocol structure.

Part 1: Capturing and Analyzing Ethernet Frames

Setup and Packet Capture

1. **Clear Browser Cache**: Before starting Wireshark, clear your browser's cache to capture only relevant traffic.
2. **Start Wireshark Capture**: Begin a new capture in Wireshark.
3. **Visit a URL**: Open this URL in your browser: http://gaia.cs.umass.edu/wireshark-labs/HTTP-ethereal-lab-file3.html.
4. **Stop Capture**: Once the page has fully loaded, stop the capture in Wireshark.

Analyzing Captured Frames

1. **Identify Ethernet Frame Structure**:
   - Locate the HTTP GET packet in the capture and examine the Ethernet frame structure.
   - **Question**:

     1. Identify the source and destination MAC addresses. Is the source MAC address from your computer, and who owns the destination MAC address?

   - **Field Analysis**: Document each field in the Ethernet header (e.g., source MAC, destination MAC, type/length field).
2. **Explore Ethernet Frame Types**:
   - **Question:**

     2. What is the hexadecimal value of the Ethernet type field, and what protocol does it correspond to? Explain why this protocol is significant for the captured traffic.

3. **Manufacturer Lookup for MAC Addresses**:
   - Use Wireshark's MAC lookup feature to identify the manufacturer of the source and destination MAC addresses.
   - **Question:**

     3. What is the manufacturer associated with the MAC address "CC:20:E8:11:22:33"? Why is manufacturer information useful in network diagnostics?

4. **Field Position Identification**:
   - **Question:**

     4. Calculate the byte offset for the ASCII "G" in the HTTP GET message from the start of the Ethernet frame. Explain how you arrived at this position.

Part 2: Comparative Protocol Analysis

1. **Ethernet and IP Layer Encapsulation**:
   - Compare an Ethernet frame to an IP packet within the captured traffic.
   - **Question:**

     5. What are the differences in encapsulation between Ethernet and IP layers? Why is encapsulation essential for effective network communication?

Part 3: ARP Packet Analysis

1. **Capture ARP Traffic**:
   - Download the Wireshark Traces and open the `NAT_home_side.pcap` file.
   - Apply the filter `arp` in Wireshark to isolate ARP packets.
2. **Analyze ARP Packet Structure**:
   - **Question:**

     6. For the first ARP packet, identify the Ethernet frame type and the destination MAC address. Is the destination a specific computer, or is it broadcasted to all devices?

3. **Explore ARP Operations**:
   - Identify the type of ARP packet and its intended operation.
   - **Question:**

     7. Describe the function of this packet. What purpose does it serve in the network, and what information does it seek or provide?

4. **ARP Header and Target MAC Analysis**:
   - **Question:**

     8. Compare the target MAC address in the ARP header with the destination MAC in the Ethernet header. Are they identical? Explain any discrepancies and their significance.

Additional Activities

1. ARP Response Timing Analysis

- **Objective**: Measure ARP request-response latency.
- **Activity**: Calculate the round-trip time between ARP requests and responses, focusing on average latency.

  - **Questions:**

    9. What is the average ARP round-trip time, and how could increased latency impact network performance?

2. MAC Address and ARP Spoofing Simulation

- **Objective**: Understand the implications of MAC and ARP spoofing in a secure, controlled lab environment.
- **Activity**: Temporarily spoof your MAC address and observe changes in Wireshark.

  - **Questions:**

    10. How does changing the MAC address affect packet analysis? What are potential security risks of MAC/ARP spoofing?

3. Dynamic ARP Cache Observation

- **Objective**: Observe the behavior of ARP cache in action.
- **Activity**: Clear your ARP cache, initiate new traffic, and observe ARP request patterns.

  - **Questions:**

    11. How frequently do ARP requests occur with and without cache? How does ARP caching impact network efficiency?